ACADEMIC COURSE DEMAND FORECASTING AND FACULTY ALLOCATION SYSTEM

Project Report

1,000	1,327	1,187
Courses forecast	Sections proposed	Sections assigned

Verified run seed: 20261022

Verification date: 30 October 2026

A prototype demonstrating data-driven academic planning using synthetic data.

Executive Summary

The Academic Course Demand Forecasting and Faculty Allocation System is a data-driven planning prototype that forecasts course demand, proposes course sections, allocates qualified faculty, assigns rooms, and reports unresolved constraints. It combines a Random Forest regression model with rule-based resource allocation, validation, audit logging, SQLite persistence, and CSV exports.

The verified run generated 1,000 course forecasts and proposed 1,327 sections. Faculty were assigned to 1,187 sections; 140 sections remained unassigned and were recorded as exceptions. Forecast evaluation against synthetic future ground truth produced an MAE of 3.902 and an RMSE of approximately 4.7672. All four automated tests passed, and the exported data-validation and plan-validation reports contained zero issues.

Important scope note: the project uses synthetic academic data. Its forecasting metrics demonstrate evaluation against generated test data and do not establish performance on real university enrollment.

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1. Problem Statement and Objectives

Academic institutions must estimate course enrollment demand and coordinate teaching resources before a term begins. Inaccurate estimates can lead to too few or too many sections, while poor faculty allocation can result in workload violations or assignments outside an instructor's qualifications. Manual planning can also make it difficult to maintain consistent records, identify anomalies, and explain unresolved constraints.

This project automates a representative academic-planning workflow. Its objectives are to:

- Generate structured synthetic datasets for academic programs, courses, faculty, skills, rooms, student plans, and historical enrollment.
- Forecast future course demand and evaluate predictions against separately stored synthetic ground truth.
- Propose course sections based on forecast demand and course capacity targets.
- Allocate faculty based on availability, qualifications, and weekly workload limits.
- Assign rooms based on capacity and room-slot occupancy.
- Validate data and planning outputs, record exceptions and audit events, and export results for analysis.

2. Technologies and System Architecture

Technology	Role in the project
Python	Core application logic and workflow orchestration.
Pandas	Data manipulation, joins, grouping, and CSV processing.
NumPy	Numerical operations and synthetic data generation.
Scikit-learn	Random Forest regression and forecast evaluation.
SQLite	Relational storage for generated and processed data.
Pytest	Automated tests for project behavior.
CSV and JSON	Portable detailed outputs and run summary.

Processing workflow

1. Generate synthetic datasets.
2. Validate input data.
3. Train and apply the demand forecasting model.
4. Calculate proposed section counts.
5. Allocate faculty and rooms.
6. Validate planning outputs and record exceptions.
7. Persist data to SQLite and export CSV/JSON reports.

3. Dataset Description

The verified run used seed 20261022. The seed makes the synthetic run reproducible for the same code and software environment.

Dataset / output	Record count
Academic programs	100
Courses	1,000
Faculty members	500
Student course-plan records	20,000
Course demand forecasts	1,000
Proposed sections	1,327
Faculty allocation records	1,327

The data is designed to exercise the pipeline. It should not be interpreted as real enrollment or staffing data from an actual university.

4. Demand Forecasting Methodology

4.1 Model

The project uses Scikit-learn's RandomForestRegressor. A Random Forest combines predictions from multiple decision trees to estimate a numerical target and can represent nonlinear relationships between historical enrollment and course characteristics.

4.2 Features and evaluation policy

The implementation uses historical enrollment lag features and course credits. The synthetic future ground-truth values are stored separately and excluded from the model features; they are used for evaluation only.

4.3 Evaluation metrics

Mean Absolute Error (MAE) measures the average absolute difference between predictions and actual values. Root Mean Squared Error (RMSE) squares errors before averaging and then takes the square root, so larger errors have greater influence. Lower values are generally better for both metrics.

Metric	Verified value	Interpretation
Model	RandomForestRegressor	Regression model for numeric demand.
Courses evaluated	1,000	One forecast per course in the run summary.
MAE	3.902	Mean absolute error in the target's units.
RMSE	4.7672	Root mean squared error in the target's units.

These values describe performance against the project's synthetic future ground truth. Real-world forecast performance would require evaluation against actual enrollment outcomes.

5. Section Planning and Resource Allocation

5.1 Section planning

Forecast demand and each course's capacity target determine the proposed number of sections. Each proposed section records its course, term, section number, forecast demand, capacity, time slot, room, and prerequisite-course information. The verified run proposed 1,327 sections.

5.2 Faculty allocation

Faculty allocation checks whether a faculty member is available, has a recorded skill for the course, and can accept the course credits without exceeding their weekly workload limit. The allocation heuristic prioritizes courses with fewer eligible faculty members. Among eligible faculty, it prefers the person with the lowest current workload utilization.

Allocation measure	Verified result
Proposed sections	1,327
Assigned sections	1,187
Unassigned sections	140
Assignment rate	89.45%
Unassigned rate	10.55%

The heuristic is a practical greedy strategy, not a guarantee of globally optimal allocation. The 140 unassigned sections are recorded as exceptions instead of being assigned to unqualified faculty or faculty without sufficient

workload capacity.

5.3 Room allocation

Rooms are considered in ascending capacity order, favoring smaller rooms that can fit a section. The system tracks room-slot occupancy to prevent a room from being assigned to more than one section in the same slot. The verified run reported 140 exceptions, corresponding to unresolved faculty assignments; the prior room-selection change had eliminated room-capacity exceptions.

The implementation does not establish that every possible room-type requirement or faculty timetable conflict is satisfied.

6. Validation and Anomaly Detection

The application includes data validation, plan validation, and five predefined anomaly scenarios. The scenarios demonstrate detection logic using deliberately constructed test cases.

Scenario	Intended detection
Faculty outside skill set	SKILL_MISMATCH
Workload exceeded	WORKLOAD_LIMIT_EXCEEDED
Section capacity too low	CAPACITY_BELOW_DEMAND
Prerequisite ordering violation	PREREQUISITE_ORDER_VIOLATION
Duplicate section assignment	DUPLICATE_SECTION_ASSIGNMENT

Validation check	Verified result
Data validation issues	0
Plan validation issues	0
Predefined anomaly scenarios	5
Automated tests passed	4 of 4

Zero reported validation issues means the implemented checks found no issues in the generated plan. It does not prove that every possible real-world scheduling constraint is enforced or detected.

7. Database, Exports, and Testing

The SQLite database is stored at `data/academic_planning.sqlite`. The verified file size was approximately 2.74 MB. The export directory contains the following ten expected files:

- `demand_forecasts.csv` - course demand predictions.
- `forecast_summary.csv` - forecast evaluation details.
- `proposed_sections.csv` - planned course sections and room assignments.
- `workload_allocations.csv` - faculty assignment outcomes.
- `exceptions.csv` - unresolved planning exceptions.
- `data_validation_issues.csv` and `plan_validation_issues.csv` - validation findings.
- `anomaly_scenarios.csv` - predefined anomaly test scenarios.
- `audit_events.csv` - allocation and exception audit records.
- `run_summary.json` - counts, metrics, seed, and ground-truth policy.

Automated testing

The command `python -m pytest -q` completed with the result **4 passed**. Additional checks confirmed 1,000 forecast rows, 1,327 proposed sections, 1,327 allocation rows, 140 exception rows, five anomaly scenarios, and zero rows in each validation-issue export.

8. Results and Discussion

Key indicator	Result
Programs	100
Courses	1,000
Faculty	500
Student plans	20,000
Forecasts	1,000
Sections proposed	1,327
Sections assigned	1,187
Sections unassigned	140
MAE	3.902
RMSE	4.7672
Data validation issues	0
Plan validation issues	0
Automated tests	4 passed

The run demonstrates an integrated workflow from synthetic data generation to forecasting, resource allocation, validation, and reporting. The allocation rate is approximately 89.45%; the remaining sections are explicitly surfaced for review. The forecasting results should be interpreted within the synthetic evaluation setup rather than as evidence of real institutional performance.

9. Limitations and Future Enhancements

Limitations

- The datasets are synthetic rather than actual university records.
- Forecast metrics are measured against generated future ground truth, not observed enrollment outcomes.
- The greedy faculty allocation heuristic does not guarantee the best possible allocation.
- The current validation does not enforce every possible prerequisite, room-type, or faculty time-conflict constraint.
- The predefined anomaly scenarios do not cover every possible operational failure.
- Performance on synthetic data may not generalize to real academic demand.

Possible future enhancements

- Evaluate forecasts using real historical enrollment data with appropriate privacy and governance controls.
- Compare Random Forest regression with alternative forecasting approaches.

- Add more comprehensive prerequisite and timetable validation.
- Add faculty time-slot conflict detection and richer room-type constraints.
- Consider advanced reassignment methods if higher allocation rates are a project requirement.
- Develop a dashboard for forecasts, workload utilization, and unresolved exceptions.

These enhancements are potential future work, not claims about functionality already implemented.

10. Conclusion

The Academic Course Demand Forecasting and Faculty Allocation System demonstrates a functioning prototype for data-driven academic planning using synthetic data. It combines course demand forecasting, section planning, faculty allocation, room assignment, validation, anomaly scenarios, audit logging, SQLite persistence, and CSV/JSON reporting.

In the verified run, the system generated 1,000 course forecasts and proposed 1,327 sections. Faculty were assigned to 1,187 sections, while 140 unresolved assignments were reported as exceptions. Forecast evaluation produced an MAE of 3.902 and an RMSE of approximately 4.7672 against synthetic future ground truth. All four automated tests passed, and the data-validation and plan-validation exports contained zero issues.

The project is suitable as a prototype and demonstration of an academic planning workflow. Evaluating it with real institutional data and validating additional operational constraints would be necessary before considering deployment in a live university environment.

End of report

Reproducibility Testing

The system was executed twice using the same random seed, **20261022**. The console summaries were identical for the reported counts and forecasting metrics. SHA-256 checksums were also calculated for the CSV files in `data/exports` after each run.

Run-to-run comparison

Metric	Run 1	Run 2
Seed	20261022	20261022
Programs	100	100
Courses	1,000	1,000
Faculty	500	500
Student plans	20,000	20,000
Forecasts	1,000	1,000
Proposed sections	1,327	1,327
Assigned sections	1,187	1,187
Unassigned sections	140	140
Exceptions	140	140
Data validation issues	0	0
Plan validation issues	0	0
MAE	3.902	3.902
RMSE	4.7671794596	4.7671794596

Checksum verification

All nine exported CSV files had identical SHA-256 checksums across the two executions: `anomaly_scenarios.csv`, `audit_events.csv`, `data_validation_issues.csv`, `demand_forecasts.csv`, `exceptions.csv`, `forecast_summary.csv`, `plan_validation_issues.csv`, `proposed_sections.csv`, and `workload_allocations.csv`. Therefore, 9 of 9 CSV outputs matched byte-for-byte between the two runs.

Interpretation and scope

Matching SHA-256 checksums indicate that the compared files are identical. The run summary contains a `created_utc` timestamp that can change between executions, so that metadata field is not expected to remain identical. The test directly verifies the nine exported CSV files and the displayed run metrics; it does not independently establish byte-for-byte reproducibility for every file outside the compared CSV set, including the SQLite database.

Conclusion: Reproducibility testing passed for the tested configuration and seed. The same seed produced matching summary counts and forecasting metrics, and all nine checked CSV exports had identical SHA-256 checksums. Because the project uses synthetic data, this test demonstrates repeatability of the implementation, not forecasting accuracy on real institutional data.